

# Scale-Optimized Geometric Acceleration of Polygonal Approximations to $\pi$

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## Abstract

This paper develops a scale-optimized geometric acceleration framework for polygonal approximations to  $\pi$ . Starting from the classical polygon sequence

$$P_N = N \sin\left(\frac{\pi}{N}\right),$$

we exploit its even-power error expansion to construct finite multi-scale operators that cancel the leading asymptotic error terms. For scale nodes  $k_m = b^m$ , the corresponding extrapolation weights are Lagrange weights at  $x = 0$  for the geometric node set  $x_m = b^{-2m}$ . This produces an accelerated hierarchy

$$P_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} P_{b^m N}$$

with error  $O(N^{-2r})$  for fixed  $b$  and  $r$ . The geometric-node structure gives a uniform stability bound

$$\sum_{m=0}^{r-1} |w_m^{(r,b)}| \leq \mathcal{S}_b, \quad \mathcal{S}_b = \prod_{\ell=1}^{\infty} \frac{1 + b^{-2\ell}}{1 - b^{-2\ell}}.$$

Numerical experiments confirm the predicted slopes through high order for the tested branches  $b = 2, 3, 6, 9, 27$ . The experiments also reveal an optimization principle: larger scale factors improve weight stability, but do not necessarily improve normalized geometric work-efficiency. In the tested data,  $27^m$  gives near-perfect weight stability, while  $6^m$  appears to provide the strongest hybrid performance among the tested scale families.

## 1 Introduction

The classical regular polygon approximation to  $\pi$  is often viewed as a slowly convergent geometric method. For a unit circle, let

$$P_N = N \sin\left(\frac{\pi}{N}\right).$$

This is the half-perimeter of the regular  $2N$ -gon inscribed in the unit circle. It satisfies

$$P_N \rightarrow \pi \quad (N \rightarrow \infty).$$

The ordinary convergence is second order:

$$P_N - \pi = O(N^{-2}).$$

However, the approximation error is not merely small; it is structured. The Taylor expansion gives

$$P_N = \pi - \frac{\pi^3}{6N^2} + \frac{\pi^5}{120N^4} - \frac{\pi^7}{5040N^6} + \cdots.$$

Thus the error has the even-power form

$$P_N = \pi + \sum_{j \geq 1} C_{2j} N^{-2j}.$$

This paper studies the consequences of that structure. By evaluating the same polygonal quantity at multiple scales and forming a suitable linear combination, one can cancel the first several even-power error modes. The result is a hierarchy of geometric acceleration operators.

The essential principle is:

Symmetry creates the even-power structure; scaling exposes it; the operator removes it.

## 2 Main Result

The central result is that polygonal approximation error is not only reduced by increasing the number of sides, but can also be cancelled across scale.

For an integer scale factor  $b \geq 2$ , define

$$P_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} P_{b^m N},$$

where the weights satisfy

$$\sum_{m=0}^{r-1} w_m^{(r,b)} = 1,$$

and

$$\sum_{m=0}^{r-1} w_m^{(r,b)} b^{-2jm} = 0, \quad j = 1, \dots, r-1.$$

Then, for fixed  $b$  and  $r$ , the accelerated polygon sequence satisfies

$$P_{N,b}^{[r]} - \pi = O(N^{-2r}).$$

Moreover, for the geometric scale nodes  $b^m$ , the extrapolation weights are uniformly stable:

$$\sum_{m=0}^{r-1} |w_m^{(r,b)}| \leq \mathcal{S}_b,$$

where

$$\mathcal{S}_b = \prod_{\ell=1}^{\infty} \frac{1 + b^{-2\ell}}{1 - b^{-2\ell}}.$$

For  $N \geq 6$ , the accelerated polygon error obeys the explicit bound

$$\left| P_{N,b}^{[r]} - \pi \right| \leq \mathcal{S}_b \frac{\pi^{2r+1}}{(2r+1)!} N^{-2r}.$$

Thus the framework combines three ingredients:

$$\boxed{\text{even-power structure} + \text{scale cancellation} + \text{uniform weight stability.}}$$

The numerical experiments further show that the choice of scale factor  $b$  is an optimization problem. Larger  $b$  improves the stability constant  $\mathcal{S}_b$ , but the best normalized geometric efficiency is not obtained simply by taking the largest tested scale factor.

### 3 The Polygonal Even-Power Expansion

The base sequence is

$$P_N = N \sin\left(\frac{\pi}{N}\right).$$

Using the sine series,

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots,$$

with  $z = \pi/N$ , we obtain

$$P_N = N \sin\left(\frac{\pi}{N}\right) = \pi - \frac{\pi^3}{6N^2} + \frac{\pi^5}{120N^4} - \frac{\pi^7}{5040N^6} + \cdots.$$

More generally,

$$P_N = \pi + \sum_{j=1}^{\infty} (-1)^j \frac{\pi^{2j+1}}{(2j+1)!} N^{-2j}.$$

**Definition 1** (Even-Power Error Structure). A sequence  $Q_N$  approximating a limit  $Q$  is said to have an even-power error structure if

$$Q_N = Q + C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \cdots.$$

The polygon sequence  $P_N$  has precisely this structure.

### 4 Scale- $b$ Acceleration Operators

Let  $b \geq 2$  be an integer scale factor. Define scale nodes

$$k_m = b^m, \quad m = 0, 1, \dots, r-1.$$

The scaled polygon approximations are

$$P_{b^m N}.$$

Let

$$x_m = k_m^{-2} = b^{-2m}.$$

The acceleration weights are chosen as the Lagrange extrapolation weights for evaluation at  $x = 0$ :

$$w_m^{(r,b)} = \prod_{\substack{i=0 \\ i \neq m}}^{r-1} \frac{-x_i}{x_m - x_i}.$$

**Definition 2** (Scale- $b$  Accelerated Polygon Approximation). For  $b \geq 2$ ,  $r \geq 1$ , define

$$P_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} P_{b^m N}.$$

The weights satisfy the moment conditions

$$\sum_{m=0}^{r-1} w_m^{(r,b)} = 1,$$

and

$$\sum_{m=0}^{r-1} w_m^{(r,b)} b^{-2jm} = 0, \quad j = 1, \dots, r-1.$$

**Theorem 1** (Scale- $b$  Error Cancellation). *Suppose*

$$Q_N = Q + C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \dots.$$

Let

$$Q_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} Q_{b^m N},$$

where the weights satisfy

$$\sum_{m=0}^{r-1} w_m^{(r,b)} = 1$$

and

$$\sum_{m=0}^{r-1} w_m^{(r,b)} b^{-2jm} = 0, \quad j = 1, \dots, r-1.$$

Then

$$Q_{N,b}^{[r]} - Q = O(N^{-2r})$$

for fixed  $b$  and  $r$ .

*Proof.* Substitute the expansion into the accelerated expression:

$$Q_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} \left[ Q + \sum_{j \geq 1} C_{2j} (b^m N)^{-2j} \right].$$

The constant term becomes

$$Q \sum_{m=0}^{r-1} w_m^{(r,b)} = Q.$$

For  $j = 1, \dots, r-1$ , the  $N^{-2j}$  term is multiplied by

$$\sum_{m=0}^{r-1} w_m^{(r,b)} b^{-2jm},$$

which is zero by construction. Hence the first possible surviving term is of order  $N^{-2r}$ . Therefore

$$Q_{N,b}^{[r]} - Q = O(N^{-2r}).$$

□

Applying this to  $Q_N = P_N$  and  $Q = \pi$  gives

$$P_{N,b}^{[r]} - \pi = O(N^{-2r}).$$

## 5 Weight Stability for Geometric Nodes

The preceding theorem explains cancellation, but not stability. For general extrapolation nodes, high-order weights may become large and numerically destructive. The geometric node structure is special.

Let

$$q = b^{-2}, \quad x_m = q^m.$$

Then

$$w_m^{(r,b)} = \prod_{\substack{i=0 \\ i \neq m}}^{r-1} \frac{-q^i}{q^m - q^i}.$$

Introduce the  $q$ -Pochhammer symbol

$$(q; q)_n = \prod_{\ell=1}^n (1 - q^\ell).$$

**Lemma 1** (Geometric-Node Weight Formula). *Let  $q = b^{-2}$  and  $x_m = q^m$ ,  $m = 0, \dots, r-1$ . Then the absolute weights may be written in the form*

$$\left| w_m^{(r,b)} \right| = \frac{q^{s(s+1)/2}}{(q; q)_m (q; q)_s}, \quad s = r - 1 - m.$$

*Proof.* Starting from

$$w_m^{(r,b)} = \prod_{\substack{i=0 \\ i \neq m}}^{r-1} \frac{-q^i}{q^m - q^i},$$

split the product into the terms  $i < m$  and  $i > m$ . The factors reduce to products of  $1 - q^\ell$ , with the powers of  $q$  collected into  $q^{s(s+1)/2}$ , where  $s = r - 1 - m$ . Taking absolute values yields the stated expression.  $\square$

**Theorem 2** (Geometric-Node Stability Bound). *Let  $q = b^{-2}$ . Then*

$$\sum_{m=0}^{r-1} \left| w_m^{(r,b)} \right| \leq \mathcal{S}_b,$$

where

$$\mathcal{S}_b = \prod_{\ell=1}^{\infty} \frac{1 + q^\ell}{1 - q^\ell} = \prod_{\ell=1}^{\infty} \frac{1 + b^{-2\ell}}{1 - b^{-2\ell}}.$$

In particular,

$$\sup_{r \geq 1} \sum_{m=0}^{r-1} \left| w_m^{(r,b)} \right| < \infty.$$

*Proof.* Using the preceding lemma,

$$\sum_{m=0}^{r-1} \left| w_m^{(r,b)} \right| = \sum_{s=0}^{r-1} \frac{q^{s(s+1)/2}}{(q; q)_{r-1-s} (q; q)_s}.$$

Since

$$(q; q)_{r-1-s} \geq (q; q)_\infty,$$

we obtain

$$\sum_{m=0}^{r-1} |w_m^{(r,b)}| \leq \frac{1}{(q; q)_\infty} \sum_{s=0}^{\infty} \frac{q^{s(s+1)/2}}{(q; q)_s}.$$

Euler's product identity gives

$$\sum_{s=0}^{\infty} \frac{q^{s(s+1)/2}}{(q; q)_s} = (-q; q)_\infty.$$

Therefore

$$\sum_{m=0}^{r-1} |w_m^{(r,b)}| \leq \frac{(-q; q)_\infty}{(q; q)_\infty} = \prod_{\ell=1}^{\infty} \frac{1+q^\ell}{1-q^\ell}.$$

Substituting  $q = b^{-2}$  gives the stated bound with  $\mathcal{S}_b$ , completing the proof.  $\square$

## 6 Analytic Remainder Bound

For the polygon sequence

$$P_N = N \sin \left( \frac{\pi}{N} \right),$$

the full expansion is

$$P_N = \pi + \sum_{j=1}^{\infty} (-1)^j \frac{\pi^{2j+1}}{(2j+1)!} N^{-2j}.$$

The scale- $b$  operator cancels the terms  $j = 1, \dots, r-1$ , leaving a controlled analytic tail.

**Theorem 3** (Scale- $b$  Remainder Bound). *Let  $N \geq 6$ ,  $b \geq 2$ , and  $r \geq 1$ . Define*

$$P_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} P_{b^m N}.$$

*Then*

$$\left| P_{N,b}^{[r]} - \pi \right| \leq \left( \sum_{m=0}^{r-1} \left| w_m^{(r,b)} \right| \right) \frac{\pi^{2r+1}}{(2r+1)!} N^{-2r}.$$

*Using Theorem 2, this gives*

$$\left| P_{N,b}^{[r]} - \pi \right| \leq \mathcal{S}_b \frac{\pi^{2r+1}}{(2r+1)!} N^{-2r},$$

*where*

$$\mathcal{S}_b = \prod_{\ell=1}^{\infty} \frac{1+b^{-2\ell}}{1-b^{-2\ell}}.$$

*Proof.* For each scaled term,

$$P_{b^m N} = \pi + \sum_{j=1}^{r-1} (-1)^j \frac{\pi^{2j+1}}{(2j+1)!} (b^m N)^{-2j} + R_r(b^m N).$$

Since  $b^m N \geq 6$ , the alternating Taylor remainder after the first  $r - 1$  nonzero terms satisfies

$$|R_r(b^m N)| \leq \frac{\pi^{2r+1}}{(2r+1)!} (b^m N)^{-2r}.$$

Applying the scale- $b$  weights cancels the first  $r - 1$  even-power terms. Therefore

$$P_{N,b}^{[r]} - \pi = \sum_{m=0}^{r-1} w_m^{(r,b)} R_r(b^m N).$$

Taking absolute values gives the sharper weighted estimate

$$\left| P_{N,b}^{[r]} - \pi \right| \leq \sum_{m=0}^{r-1} \left| w_m^{(r,b)} \right| \frac{\pi^{2r+1}}{(2r+1)!} (b^m N)^{-2r}.$$

Since  $b^m \geq 1$ , we have

$$(b^m N)^{-2r} \leq N^{-2r}.$$

Hence

$$\left| P_{N,b}^{[r]} - \pi \right| \leq \left( \sum_{m=0}^{r-1} \left| w_m^{(r,b)} \right| \right) \frac{\pi^{2r+1}}{(2r+1)!} N^{-2r}.$$

The final bound follows from the stability theorem.  $\square$

*Remark 1* (Meaning of the Bound). The estimate shows that the accelerated hierarchy is not only formally high-order. The remaining analytic tail is controlled by two explicit factors: the geometric-node stability constant  $\mathcal{S}_b$ , and the factorial decay

$$\frac{1}{(2r+1)!}.$$

Thus the scale- $b$  construction combines asymptotic cancellation with a quantified analytic remainder.

## 7 Constructible Chord Recursion

The analytic formula

$$P_N = N \sin\left(\frac{\pi}{N}\right)$$

is useful for analysis, but a geometric interpretation should not depend on direct trigonometric evaluation.

Let  $c_N$  denote the chord length of the regular  $N$ -gon in the unit circle:

$$c_N = 2 \sin\left(\frac{\pi}{N}\right).$$

Then

$$P_N = \frac{N}{2} c_N.$$

Starting from the regular hexagon,

$$c_6 = 1.$$

The stable bisection recurrence is

$$c_{2N} = \frac{c_N}{\sqrt{2 + \sqrt{4 - c_N^2}}}.$$

This generates the sequence

$$6, 12, 24, 48, \dots$$

using only rational operations and square roots.

For tripling, the chord relation is

$$c_N = 3c_{3N} - c_{3N}^3.$$

Thus  $u = c_{3N}$  is the small positive root of

$$u^3 - 3u + c_N = 0.$$

The numerical experiments using tripling, power-of-6, power-of-9, and power-of-27 used this cubic inversion as an architecture test. A fully compass-and-straightedge version of the tripling branch would require replacing this cubic inversion with a constructible trisection/refinement approximation.

*Remark 2* (Constructibility Status). The  $2^m$  branch is strictly chord-recursive by nested square roots. The branches involving factors of 3 are numerical architecture tests unless the cubic inversion is replaced by a constructible trisection/refinement mechanism.

## 8 Algorithmic Form of the Scale- $b$ Operator

For clarity, we summarize the computational form of the scale- $b$  geometric acceleration operator.

**Definition 3** (Scale- $b$  Polygon Acceleration Algorithm). Fix  $b \geq 2$ ,  $r \geq 1$ , and a base polygon parameter  $N$ . The scale- $b$  accelerated polygon approximation is computed as follows.

1. Form the scale nodes

$$k_m = b^m, \quad m = 0, 1, \dots, r-1.$$

2. Define the extrapolation nodes

$$x_m = k_m^{-2} = b^{-2m}.$$

3. Compute the Lagrange extrapolation weights at  $x = 0$ :

$$w_m^{(r,b)} = \prod_{\substack{i=0 \\ i \neq m}}^{r-1} \frac{-x_i}{x_m - x_i}.$$

4. Evaluate the polygonal approximations

$$P_{b^m N}, \quad m = 0, 1, \dots, r-1.$$

5. Form the accelerated approximation

$$P_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} P_{b^m N}.$$

The defining property of the weights is equivalently expressed by the moment conditions

$$\sum_{m=0}^{r-1} w_m^{(r,b)} = 1,$$

and

$$\sum_{m=0}^{r-1} w_m^{(r,b)} b^{-2jm} = 0, \quad j = 1, \dots, r-1.$$

Thus the operator preserves the limiting value while cancelling the first  $r-1$  even-power error modes.

*Remark 3* (Geometric Versus Analytic Evaluation). The polygonal quantities  $P_{b^m N}$  may be evaluated analytically as

$$P_{b^m N} = b^m N \sin\left(\frac{\pi}{b^m N}\right),$$

or geometrically from chord lengths as

$$P_{b^m N} = \frac{b^m N}{2} c_{b^m N}.$$

The analytic form is useful for proof and validation, while the chord form is the geometric form of the construction.

For the purely bisectional branch  $b = 2$ , the required chord lengths are generated from

$$c_6 = 1, \quad c_{2N} = \frac{c_N}{\sqrt{2 + \sqrt{4 - c_N^2}}},$$

using only rational operations and square roots. For branches involving factors of 3, the experiments used the chord tripling relation

$$c_N = 3c_{3N} - c_{3N}^3.$$

These tripling branches test the scale architecture, while strict compass-and-straightedge constructibility remains tied to the open tripling/refinement problem discussed below.

## 9 Proof Dependencies and Status of the Claims

The framework separates into three levels of justification.

1. **Exact cancellation.** The cancellation of the first  $r-1$  even-power terms follows algebraically from the moment conditions

$$\sum_{m=0}^{r-1} w_m^{(r,b)} b^{-2jm} = 0, \quad j = 1, \dots, r-1.$$

This part is exact.

2. **Uniform weight stability.** For geometric scale nodes  $x_m = b^{-2m}$ , the extrapolation weights satisfy the uniform bound

$$\sum_{m=0}^{r-1} |w_m^{(r,b)}| \leq \mathcal{S}_b,$$

where

$$\mathcal{S}_b = \prod_{\ell=1}^{\infty} \frac{1 + b^{-2\ell}}{1 - b^{-2\ell}}.$$

This part follows from the geometric-node weight formula and Euler's product identity.

3. **Constructive realization.** For  $b = 2$ , the required polygonal values are generated by nested square roots from the regular hexagon. For branches involving factors of 3, the numerical architecture tests use cubic inversion to generate exact tripled chord values. Section 15 shows that the elementary seed

$$\tilde{c}_{3N}^{(0)} = \frac{c_N}{3}$$

is constructible and has chord error  $O(N^{-3})$ . Thus both the elementary seed and the cubic refinement step are constructible. The remaining distinction is that the numerical architecture tests use exact cubic inversion for  $c_{3N}$ , while the fully geometric hierarchy uses the constructible seed and finitely many cubic refinements.

Accordingly, the rigorously established part of the framework is the multi-scale cancellation and stability mechanism. The numerical experiments support the scale-optimization claims and identify promising branches, while Section 15 shows how exact cubic inversion may be replaced by a constructible seed-and-refinement hierarchy.

## 10 Numerical Protocol and Results

The experiments tested scale families

$$b = 2, 3, 6, 9, 27.$$

For each branch, the accelerated approximation was computed as

$$P_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} P_{b^m N},$$

where the weights satisfy

$$\sum_{m=0}^{r-1} w_m^{(r,b)} = 1,$$

and

$$\sum_{m=0}^{r-1} w_m^{(r,b)} b^{-2jm} = 0, \quad j = 1, \dots, r-1.$$

The purpose of the computations was not merely to evaluate  $\pi$ , but to test the structural predictions of the scale-accelerated geometric framework:

$$P_{N,b}^{[r]} - \pi = O(N^{-2r}),$$

the boundedness of the extrapolation weights,

$$\sum_{m=0}^{r-1} |w_m^{(r,b)}| \leq \mathcal{S}_b,$$

and the dependence of geometric work-efficiency on the scale factor  $b$ .

The observed convergence slopes were obtained by fitting

$$\log_{10} |P_{N,b}^{[r]} - \pi|$$

against

$$\log_{10} N.$$

Rows too close to the working-precision floor were excluded from the slope fits. This avoids contaminating the regression with numerical cancellation noise after the theoretical error has fallen below the reliable precision window.

## 10.1 Reproducibility Protocol

For each tested scale family  $b^m$ , the computation used the following steps.

1. Fix the working precision and the maximum acceleration order  $r$ .
2. Generate or evaluate the required polygonal quantities  $P_{b^m N}$ .
3. Compute the Lagrange extrapolation weights from the nodes

$$x_m = b^{-2m}.$$

4. Form the accelerated value

$$P_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} P_{b^m N}.$$

5. Measure the error against the reference value of  $\pi$ :

$$E_{N,b}^{[r]} = P_{N,b}^{[r]} - \pi.$$

6. Fit

$$\log_{10} |E_{N,b}^{[r]}|$$

against

$$\log_{10} N$$

over rows that remain above the working-precision floor.

Rows near the precision floor were excluded because the true error may become smaller than the reliable numerical resolution. In that regime, subtracting  $\pi$  from the accelerated approximation can produce cancellation noise that corrupts the fitted slope. The exclusion window was therefore used only to prevent numerical artifacts, not to improve the theoretical prediction.

The following table summarizes the precision and maximum tested order for each branch.

| Scale family | Precision   | Max $r$ | Expected slope | Verified behavior         |
|--------------|-------------|---------|----------------|---------------------------|
| $2^m$        | 300 digits  | 12      | $-2r$          | verified through $r = 12$ |
| $3^m$        | 800 digits  | 24      | $-2r$          | verified through $r = 24$ |
| $6^m$        | 1000 digits | 24      | $-2r$          | verified through $r = 24$ |
| $9^m$        | 1200 digits | 24      | $-2r$          | verified through $r = 24$ |
| $27^m$       | 1500 digits | 24      | $-2r$          | verified through $r = 24$ |

Table 1: Numerical protocol for the tested scale families. The predicted slope for the  $r$ -scale accelerated operator is  $-2r$ .

The stability constant for the geometric scale family  $b^m$  is

$$\mathcal{S}_b = \prod_{\ell=1}^{\infty} \frac{1 + b^{-2\ell}}{1 - b^{-2\ell}}.$$

| Scale family | $q = b^{-2}$ | Stability constant $\mathcal{S}_b$ | Interpretation                         |
|--------------|--------------|------------------------------------|----------------------------------------|
| $2^m$        | 1/4          | 1.96926004...                      | strict nested-radical bisection branch |
| $3^m$        | 1/9          | 1.285210586...                     | pure tripling branch                   |
| $6^m$        | 1/36         | 1.058822194...                     | hybrid tripling-bisection branch       |
| $9^m$        | 1/81         | 1.025316406...                     | double-tripling branch                 |
| $27^m$       | 1/729        | 1.002751031...                     | triple-tripling branch                 |

Table 2: Observed/theoretical stability constants for geometric scale families. As  $b$  increases, the weight amplification approaches 1.

Representative high-order slope checks are:

| Branch | $r$ | Observed slope |
|--------|-----|----------------|
| $3^m$  | 24  | -47.9999965... |
| $6^m$  | 24  | -47.9999966... |
| $9^m$  | 24  | -47.9999967... |
| $27^m$ | 24  | -47.9999953... |

Thus the experiments support the predicted hierarchy

$$P_{N,b}^{[r]} - \pi = O(N^{-2r})$$

for each tested scale family.

| Scale family | Best observed efficiency       | Normalized interpretation             |
|--------------|--------------------------------|---------------------------------------|
| $2^m$        | 12.68 digits / bisection level | strict constructible branch           |
| $3^m$        | 17.69 digits / tripling level  | strongest pure tripling branch tested |
| $6^m$        | 26.30 digits / power-6 level   | strongest hybrid branch tested        |
| $9^m$        | 31.26 digits / power-9 level   | 15.63 digits / tripling substep       |
| $27^m$       | 43.19 digits / power-27 level  | 14.40 digits / tripling substep       |

Table 3: Best observed work-efficiency values. Larger scale factors improve weight stability, but the normalized geometric efficiency does not increase monotonically. The  $27^m$  branch has the smallest weight amplification, but its normalized tripling efficiency is lower than that of the  $3^m$  branch.

## 10.2 Interpretation of the Numerical Data

The numerical data support three separate conclusions.

First, the measured slopes agree with the predicted cancellation order. For an  $r$ -scale operator, the expected slope is

$$-2r.$$

The tested branches agree with this prediction through the reported reliability windows, including the high-order cases  $r = 24$  for the branches involving scale factors 3, 6, 9, and 27.

Second, the weight stability constants decrease as the scale factor increases. For geometric scale nodes  $b^m$ , the stability constant is

$$\mathcal{S}_b = \prod_{\ell=1}^{\infty} \frac{1 + b^{-2\ell}}{1 - b^{-2\ell}}.$$

Thus increasing  $b$  moves the extrapolation weights closer to an ideal unit-norm operator. This explains why the  $27^m$  branch has the smallest observed weight amplification among the tested families.

Third, the work-efficiency data show that stability alone does not determine the optimal scale family. Although the  $27^m$  branch has the smallest weight constant, it does not give the best normalized efficiency per elementary tripling substep. The  $3^m$  branch remains stronger among the pure tripling families, while  $6^m$  gives the strongest tested hybrid behavior.

Therefore, the experiments indicate that the optimal scale architecture is a balance between three quantities:

$$\boxed{\text{scale separation} \quad + \quad \text{weight stability} \quad + \quad \text{elementary construction cost.}}$$

The numerical results therefore support a non-monotone optimization principle. Increasing the scale factor improves the stability constant, but does not necessarily improve the normalized geometric efficiency. Among the tested families,  $3^m$  is the strongest pure tripling branch, while  $6^m$  is the strongest hybrid branch.

## 11 Optimization Principle

The experiments reveal a tradeoff. Increasing  $b$  improves the stability constant:

$$\mathcal{S}_b \rightarrow 1 \quad (b \rightarrow \infty).$$

However, increasing  $b$  also increases the geometric work required to reach the scales  $b^m N$ .

*Remark 4 (Stability Versus Efficiency).* Large scale ratios reduce weight amplification, but may waste geometric substeps. The  $27^m$  branch has near-perfect stability, but lower normalized tripling efficiency than  $3^m$  and  $9^m$ . Thus the optimal scale factor is not the largest scale factor. It is the one that best balances scale jump, weight stability, and geometric work.

The tested data suggest:

$$\boxed{3^m \text{ is the strongest pure tripling branch tested,}}$$

while

$$\boxed{6^m \text{ is the strongest hybrid branch tested.}}$$

## 12 Limitations and Scope

The results in this paper should be interpreted as evidence for a scale-optimized geometric acceleration hierarchy, not as a claim of a new fastest numerical algorithm for computing  $\pi$ .

The  $2^m$  branch is fully chord-recursive from the regular hexagon using only rational operations and square roots. In contrast, the branches involving factors of 3 use numerical cubic inversion to

generate the tripled chord values. These branches therefore test the scale architecture and operator stability. The elementary seed  $c_N/3$  and the Baker cubic refinement step are constructible. A fully strict compass-and-straightedge version of the tripling and hybrid branches should therefore be formulated using the seed and refinement hierarchy rather than exact cubic inversion.

A fully constructible version of the tripling and hybrid branches is obtained by replacing the cubic inversion

$$u^3 - 3u + c_N = 0$$

with the constructible seed-and-refinement hierarchy developed in Section 15, whose perturbation error is small enough to preserve the observed accelerated hierarchy after finitely many refinements.

Accordingly, the main contribution is the identification of a stable multi-scale geometric cancellation structure:

$$P_{N,b}^{[r]} - \pi = O(N^{-2r}),$$

together with a uniform weight-stability mechanism for geometric scale nodes. The numerical experiments indicate that scale choice matters: larger scale ratios improve the weight constant, but normalized geometric efficiency is not monotone in the scale factor.

## 13 Comparison with AGM-Based Algorithms

The Gauss–Legendre algorithm remains superior as a practical high-speed numerical algorithm for computing  $\pi$ . It is based on the arithmetic-geometric mean and exhibits quadratic convergence, roughly doubling the number of correct digits at each iteration.

For background on AGM-based methods for  $\pi$ , see Borwein and Borwein [3].

The present framework is different. It is not primarily an arithmetic iteration, but a scale-indexed geometric cancellation hierarchy. Its purpose is to expose and remove the structured even-power error modes in polygonal geometry.

Thus the appropriate comparison is not simply numerical speed. The geometric hierarchy shows that regular polygon approximations possess a stable, high-order cancellation structure. This is a geometric acceleration principle rather than a replacement for AGM methods.

## 14 Benefits and Interpretation

The primary benefit of the scale-optimized polygon framework is not that it immediately replaces AGM-based algorithms as a practical method for computing  $\pi$ . Rather, its significance is that it exposes a stable geometric error-cancellation structure inside regular polygon approximation.

Classically, polygonal approximation improves by refinement: one increases the number of sides and obtains a better approximation to the circle. The present framework shows that this is not the only mechanism available. Because the polygonal error has an even-power structure,

$$P_N - \pi = C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \dots,$$

scaled polygonal values may be combined so that the leading error modes cancel. Thus the circle is approached not only by refinement, but by cancellation of refinement error.

This produces a hierarchy of accelerated geometric approximations satisfying

$$P_{N,b}^{[r]} - \pi = O(N^{-2r}),$$

with the explicit bound

$$\left| P_{N,b}^{[r]} - \pi \right| \leq \mathcal{S}_b \frac{\pi^{2r+1}}{(2r+1)!} N^{-2r}.$$

The role of the stability constant  $\mathcal{S}_b$  is important: it shows that the high-order cancellation is not merely formal, since the geometric scale weights remain uniformly controlled.

A second benefit is that the method turns scale choice into a genuine geometric optimization problem. Larger scale factors improve the stability constant, but the experiments show that normalized work-efficiency is not monotone in the scale factor. Thus the best architecture is not obtained simply by taking the largest scale jump. It is obtained by balancing scale separation, weight stability, and elementary construction cost.

Finally, the framework provides a bridge between constructible polygon geometry and high-order numerical acceleration. The  $2^m$  branch is generated from the regular hexagon by nested square roots, while the branches involving factors of 3 identify a natural role for trisection or tripling-refinement mechanisms. In this sense, the method strengthens the geometric interpretation of numerical approximation: the polygon is not just a crude approximation to the circle, but a scale-indexed carrier of removable error structure.

## 15 Constructible Tripling Replacement

The scale-3 and scale-6 branches are the most important constructible replacement targets. In the numerical architecture tests, tripled chord values are obtained by solving

$$u^3 - 3u + c_N = 0,$$

where  $u = c_{3N}$ . This cubic inversion is not generally a compass-and-straightedge operation. Therefore, a fully constructible version of the tripling and hybrid branches requires replacing this cubic inversion with a constructible trisection or refinement mechanism.

The preceding numerical results suggest that such a replacement would be especially valuable. The  $3^m$  family is the strongest pure tripling branch tested, while the  $6^m$  family is the strongest hybrid branch tested. Since the  $6^m$  branch consists of a tripling component and an exact bisection component, its remaining constructibility obstruction is again the tripling step.

The purpose of this section is to convert the constructible tripling question into a quantitative perturbation problem. Rather than asking only whether  $c_{3N}$  can be approximated constructibly, we ask how accurately it must be approximated in order for the accelerated hierarchy to survive.

The analysis below shows that the elementary seed  $c_N/3$  supplies a constructible approximation with algebraic accuracy exponent  $\beta = 3$ , and that the Baker cubic refinement step is itself compass-and-straightedge constructible. Thus exact cubic inversion can be replaced, for the accelerated hierarchy, by a fully geometric seed-and-refinement procedure.

### 15.1 Perturbation Requirement for Approximate Tripling

The tripling branches require the chord value  $c_{3N}$ . In the numerical architecture tests, this value is obtained by solving

$$u^3 - 3u + c_N = 0,$$

where  $u = c_{3N}$ . A constructible replacement will generally produce an approximation

$$\tilde{c}_{3N} = c_{3N} + \delta_N.$$

The question is how small the chord perturbation  $\delta_N$  must be in order for the accelerated hierarchy to remain valid.

Since

$$P_N = \frac{N}{2} c_N,$$

a chord perturbation at scale  $M$  produces a polygonal perturbation

$$\tilde{P}_M - P_M = \frac{M}{2} \delta_M.$$

For the scale-3 accelerated approximation

$$P_{N,3}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,3)} P_{3^m N},$$

suppose the computed chord value at scale  $3^m N$  is perturbed by

$$\tilde{c}_{3^m N} = c_{3^m N} + \delta_{3^m N}.$$

Then the induced perturbation in the accelerated approximation is

$$\tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,3)} \frac{3^m N}{2} \delta_{3^m N}.$$

Therefore

$$\left| \tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} \right| \leq \frac{N}{2} \sum_{m=0}^{r-1} |w_m^{(r,3)}| 3^m |\delta_{3^m N}|.$$

**Theorem 4** (Sufficient Tripling Perturbation Condition). *Let*

$$P_{N,3}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,3)} P_{3^m N}$$

*be the exact scale-3 accelerated polygon approximation. Suppose the tripled chord values used in the computation are perturbed by*

$$\tilde{c}_{3^m N} = c_{3^m N} + \delta_{3^m N}.$$

*If*

$$\frac{N}{2} \sum_{m=0}^{r-1} |w_m^{(r,3)}| 3^m |\delta_{3^m N}| \leq K \mathcal{S}_3 \frac{\pi^{2r+1}}{(2r+1)!} N^{-2r}$$

*for some fixed constant  $K$ , then the perturbation introduced by the approximate tripling construction is no larger than the analytic accelerated remainder scale. In particular,*

$$\left| \tilde{P}_{N,3}^{[r]} - \pi \right| = O(N^{-2r})$$

*provided the unperturbed scale-3 hierarchy satisfies*

$$\left| P_{N,3}^{[r]} - \pi \right| = O(N^{-2r}).$$

*Proof.* The accelerated approximation computed from perturbed chord values differs from the exact accelerated approximation by

$$\tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,3)} \frac{3^m N}{2} \delta_{3^m N}.$$

Taking absolute values gives

$$\left| \tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} \right| \leq \frac{N}{2} \sum_{m=0}^{r-1} |w_m^{(r,3)}| 3^m |\delta_{3^m N}|.$$

By the assumed perturbation condition, this is bounded by the same order as the analytic accelerated remainder:

$$O(N^{-2r}).$$

Hence

$$\tilde{P}_{N,3}^{[r]} - \pi = \left( P_{N,3}^{[r]} - \pi \right) + \left( \tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} \right) = O(N^{-2r}).$$

□

This theorem converts the constructible tripling problem into an explicit error-tolerance problem. It is not enough to approximate  $c_{3N}$  qualitatively; the chord error must be small enough that, after multiplication by the polygon scale  $3^m N/2$  and by the extrapolation weights, it remains below the analytic accelerated remainder. Thus the open problem becomes quantitative:

Can the constructible trisection refinement produce  $|\delta_{3^m N}|$  small enough to satisfy the perturbation bound?

**Corollary 1** (Scale-Law Condition for Tripling Perturbations). *Let  $M = 3^m N$ , and suppose the chord perturbations obey the scale law*

$$|\delta_M| \leq AM^{-\alpha}$$

*for constants  $A > 0$  and  $\alpha > 0$ . Then the perturbation contribution to the scale-3 accelerated approximation satisfies*

$$\left| \tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} \right| \leq \frac{A}{2} N^{1-\alpha} \sum_{m=0}^{r-1} |w_m^{(r,3)}| 3^{m(1-\alpha)}.$$

*In particular, if*

$$\alpha \geq 2r + 1,$$

*then the perturbation contribution is*

$$O(N^{-2r}),$$

*and therefore does not destroy the  $r$ -scale accelerated hierarchy.*

*Proof.* From the perturbation estimate,

$$\left| \tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} \right| \leq \frac{N}{2} \sum_{m=0}^{r-1} |w_m^{(r,3)}| 3^m |\delta_{3^m N}|.$$

Using the assumed scale law with  $M = 3^m N$ ,

$$|\delta_{3^m N}| \leq A(3^m N)^{-\alpha}.$$

Hence

$$\left| \tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} \right| \leq \frac{N}{2} \sum_{m=0}^{r-1} |w_m^{(r,3)}| 3^m A(3^m N)^{-\alpha}.$$

Simplifying gives

$$\left| \tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} \right| \leq \frac{A}{2} N^{1-\alpha} \sum_{m=0}^{r-1} |w_m^{(r,3)}| 3^{m(1-\alpha)}.$$

For fixed  $r$ , the finite weighted sum is bounded independently of  $N$ . If  $\alpha \geq 2r + 1$ , then

$$N^{1-\alpha} \leq N^{-2r}.$$

Therefore the perturbation contribution is  $O(N^{-2r})$ .  $\square$

*Remark 5 (Tripling Accuracy Target).* The corollary gives a concrete accuracy target for a constructible tripling replacement. To preserve an  $r$ -scale accelerated hierarchy, the chord tripling error should satisfy

$$|\delta_M| = O(M^{-(2r+1)}).$$

Thus higher-order acceleration requires correspondingly higher-order accuracy in the tripling construction. This converts the constructible tripling problem into a quantitative approximation problem.

*Remark 6 (Connection with Cubic Refinement).* The Baker trisection refinement has the local contraction form

$$e_+ = \frac{1}{24} e^3 + O(e^5),$$

where  $e$  denotes the angular error. If the initial constructible tripling approximation has angular error  $e_0(M) = O(M^{-\beta})$ , then after one cubic refinement the error becomes

$$e_1(M) = O(M^{-3\beta}),$$

and after  $s$  cubic refinements,

$$e_s(M) = O(M^{-3^s \beta}).$$

Since chord error is locally proportional to angular error, the same order law applies to  $\delta_M$ . Therefore a sufficient condition for preserving the  $r$ -scale hierarchy is

$$3^s \beta \geq 2r + 1.$$

Equivalently,

$$s \geq \left\lceil \log_3 \left( \frac{2r+1}{\beta} \right) \right\rceil.$$

This suggests that only logarithmically many cubic refinements are needed, in principle, to meet the tripling perturbation tolerance for order  $r$ .

**Proposition 1** (Constructible Linear Tripling Seed). *Let*

$$c_N = 2 \sin \left( \frac{\pi}{N} \right)$$

*be the chord length corresponding to the angle  $\pi/N$ . Define the linear tripling seed by*

$$\tilde{c}_{3N}^{(0)} = \frac{c_N}{3}.$$

*Then this seed is compass-and-straightedge constructible from  $c_N$ , and*

$$\tilde{c}_{3N}^{(0)} - c_{3N} = -\frac{8\pi^3}{81}N^{-3} + O(N^{-5}).$$

*In particular,*

$$\left| \tilde{c}_{3N}^{(0)} - c_{3N} \right| = O(N^{-3}).$$

*Thus the seed has algebraic accuracy exponent*

$$\beta = 3.$$

*Proof.* The segment  $c_N/3$  is constructible from  $c_N$ , since a given segment may be divided into three equal parts using parallel-line construction.

Let

$$x = \frac{\pi}{N}.$$

Then

$$c_N = 2 \sin x, \quad c_{3N} = 2 \sin \left( \frac{x}{3} \right).$$

The proposed seed is

$$\tilde{c}_{3N}^{(0)} = \frac{c_N}{3} = \frac{2}{3} \sin x.$$

Using the Taylor expansion of sine,

$$\frac{2}{3} \sin x = \frac{2}{3} \left( x - \frac{x^3}{6} + O(x^5) \right) = \frac{2x}{3} - \frac{x^3}{9} + O(x^5).$$

On the other hand,

$$2 \sin \left( \frac{x}{3} \right) = 2 \left( \frac{x}{3} - \frac{x^3}{162} + O(x^5) \right) = \frac{2x}{3} - \frac{x^3}{81} + O(x^5).$$

Subtracting gives

$$\tilde{c}_{3N}^{(0)} - c_{3N} = \left( \frac{2x}{3} - \frac{x^3}{9} + O(x^5) \right) - \left( \frac{2x}{3} - \frac{x^3}{81} + O(x^5) \right).$$

Therefore

$$\tilde{c}_{3N}^{(0)} - c_{3N} = -\frac{8}{81}x^3 + O(x^5).$$

Since  $x = \pi/N$ , we obtain

$$\tilde{c}_{3N}^{(0)} - c_{3N} = -\frac{8\pi^3}{81}N^{-3} + O(N^{-5}).$$

Hence

$$\left| \tilde{c}_{3N}^{(0)} - c_{3N} \right| = O(N^{-3}),$$

so the seed exponent is  $\beta = 3$ . □

**Lemma 2** (Angular Error of the Linear Seed). *Let*

$$x = \frac{\pi}{N}, \quad y = \frac{x}{3}.$$

*Let the angular seed  $\tilde{y}_0$  be defined by*

$$2 \sin \tilde{y}_0 = \frac{2}{3} \sin x.$$

*Equivalently,*

$$\tilde{y}_0 = \arcsin \left( \frac{1}{3} \sin x \right).$$

*Then*

$$\tilde{y}_0 - y = -\frac{4}{81}x^3 + O(x^5).$$

*Consequently,*

$$\tilde{y}_0 - \frac{x}{3} = O(N^{-3}).$$

*Proof.* Using

$$\sin x = x - \frac{x^3}{6} + O(x^5),$$

we have

$$\frac{1}{3} \sin x = \frac{x}{3} - \frac{x^3}{18} + O(x^5).$$

Since

$$\arcsin z = z + \frac{z^3}{6} + O(z^5),$$

with

$$z = \frac{x}{3} - \frac{x^3}{18} + O(x^5),$$

we obtain

$$\arcsin z = \frac{x}{3} - \frac{x^3}{18} + \frac{1}{6} \left( \frac{x}{3} \right)^3 + O(x^5).$$

Thus

$$\tilde{y}_0 = \frac{x}{3} - \frac{x^3}{18} + \frac{x^3}{162} + O(x^5) = \frac{x}{3} - \frac{4x^3}{81} + O(x^5).$$

Therefore

$$\tilde{y}_0 - \frac{x}{3} = -\frac{4}{81}x^3 + O(x^5).$$

Since  $x = \pi/N$ , the angular seed error is  $O(N^{-3})$ . □

**Theorem 5** (Linear Seed with Baker Cubic Refinement). *By Theorem 6, the constructible refinement map has local angular error law*

$$e_+ = -\frac{7}{384}e^3 + O(e^5).$$

*Let the initial angular seed be induced by the constructible chord seed*

$$\tilde{c}_{3N}^{(0)} = \frac{c_N}{3}.$$

Then after  $s$  cubic refinements, the angular and chord errors satisfy

$$e_s = O(N^{-3^{s+1}}),$$

and

$$\delta_N^{(s)} = O(N^{-3^{s+1}}).$$

Consequently, the refined linear seed preserves an  $r$ -scale accelerated tripling hierarchy whenever

$$3^{s+1} \geq 2r + 1.$$

Equivalently,

$$s \geq \lceil \log_3(2r + 1) \rceil - 1.$$

*Proof.* By the preceding lemma, the initial angular error satisfies

$$e_0 = O(N^{-3}).$$

The refinement law gives

$$e_1 = -\frac{7}{384}e_0^3 + O(e_0^5) = O(N^{-9}).$$

Applying the same cubic contraction repeatedly gives

$$e_s = O(N^{-3 \cdot 3^s}) = O(N^{-3^{s+1}}).$$

For small angles, chord error is locally proportional to angular error, because

$$\frac{d}{dy} (2 \sin y) = 2 \cos y,$$

and  $2 \cos y$  remains bounded and nonzero near  $y = 0$ . Therefore

$$\delta_N^{(s)} = O(N^{-3^{s+1}}).$$

The perturbation theory requires chord accuracy

$$\delta_N^{(s)} = O(N^{-(2r+1)}).$$

Thus it is sufficient that

$$3^{s+1} \geq 2r + 1.$$

Solving for  $s$  gives

$$s \geq \lceil \log_3(2r + 1) \rceil - 1.$$

□

**Theorem 6** (Constructibility of the Baker Cubic Refinement). *By Theorem 6, the constructible refinement map has local angular error law*

$$e_+ = -\frac{7}{384}e^3 + O(e^5).$$

Let the initial angular seed be induced by the constructible chord seed

$$\tilde{c}_{3N}^{(0)} = \frac{c_N}{3}.$$

Then after  $s$  cubic refinements, the angular and chord errors satisfy

$$e_s = O(N^{-3^{s+1}}),$$

and

$$\delta_N^{(s)} = O(N^{-3^{s+1}}).$$

Consequently, the refined linear seed preserves an  $r$ -scale accelerated tripling hierarchy whenever

$$3^{s+1} \geq 2r + 1.$$

Equivalently,

$$s \geq \lceil \log_3(2r + 1) \rceil - 1.$$

*Proof.* By the preceding lemma, the initial angular error satisfies

$$e_0 = O(N^{-3}).$$

The refinement law gives

$$e_1 = -\frac{7}{384}e_0^3 + O(e_0^5) = O(N^{-9}).$$

Applying the same cubic contraction repeatedly gives

$$e_s = O(N^{-3 \cdot 3^s}) = O(N^{-3^{s+1}}).$$

For small angles, chord error is locally proportional to angular error, because

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and  $2 \cos y$  remains bounded and nonzero near  $y = 0$ . Therefore

$$\delta_N^{(s)} = O(N^{-3^{s+1}}).$$

The perturbation theory requires chord accuracy

$$\delta_N^{(s)} = O(N^{-(2r+1)}).$$

Thus it is sufficient that

$$3^{s+1} \geq 2r + 1.$$

Solving for  $s$  gives

$$s \geq \lceil \log_3(2r + 1) \rceil - 1.$$

□

**Corollary 2** (Three Refinements Suffice for  $r = 24$ ). *For  $r = 24$ , the required perturbation exponent is*

$$2r + 1 = 49.$$

*The linear seed has exponent  $\beta = 3$ . After  $s$  cubic refinements the effective exponent is*

$$3^{s+1}.$$

*Since*

$$3^3 = 27 < 49, \quad 3^4 = 81 \geq 49,$$

*it follows that*

$$s = 3$$

*cubic refinements are sufficient to preserve the  $r = 24$  accelerated hierarchy.*

**Theorem 7** (Conditional Constructible Tripling Preservation). *Suppose there exists a constructible tripling approximation producing chord values*

$$\tilde{c}_M = c_M + \delta_M$$

*with initial scale-law accuracy*

$$|\delta_M| \leq AM^{-\beta}$$

*for constants  $A > 0$  and  $\beta > 0$ . Suppose further that a constructible refinement step improves the local chord error cubically, so that after  $s$  refinements the chord perturbation satisfies*

$$|\delta_M^{(s)}| \leq A_s M^{-3^s \beta}$$

*for some constant  $A_s > 0$ .*

*Then, for any fixed acceleration order  $r$ , if*

$$3^s \beta \geq 2r + 1,$$

*the perturbed scale-3 accelerated approximation satisfies*

$$\tilde{P}_{N,3}^{[r]} - \pi = O(N^{-2r}).$$

*Proof.* By assumption, after  $s$  refinements the chord perturbation obeys

$$|\delta_M^{(s)}| \leq A_s M^{-3^s \beta}.$$

This is the scale-law condition of Corollary 1 with

$$\alpha = 3^s \beta.$$

If

$$3^s \beta \geq 2r + 1,$$

then the corollary implies that the perturbation contribution to the accelerated approximation is  $O(N^{-2r})$ . Since the exact scale-3 accelerated hierarchy also satisfies

$$P_{N,3}^{[r]} - \pi = O(N^{-2r}),$$

the perturbed approximation satisfies

$$\tilde{P}_{N,3}^{[r]} - \pi = \left( P_{N,3}^{[r]} - \pi \right) + \left( \tilde{P}_{N,3}^{[r]} - P_{N,3}^{[r]} \right) = O(N^{-2r}).$$

□

**Corollary 3** (Hybrid Scale-6 Preservation). *Consider the hybrid scale family*

$$k_m = 6^m.$$

*Suppose the bisection component is generated exactly by the nested-radical recursion*

$$c_{2N} = \frac{c_N}{\sqrt{2 + \sqrt{4 - c_N^2}}},$$

and suppose the tripling component is replaced by an approximate constructible tripling rule whose chord error after  $s$  refinements satisfies

$$|\delta_M^{(s)}| \leq A_s M^{-3^s \beta}.$$

If

$$3^s \beta \geq 2r + 1,$$

then the perturbed hybrid scale-6 accelerated approximation satisfies

$$\tilde{P}_{N,6}^{[r]} - \pi = O(N^{-2r}).$$

*Proof.* The scale-6 branch is obtained by combining a tripling step with an exact bisection step. The bisection step introduces no approximation error beyond the exact nested-radical chord construction. Therefore the only perturbation source is the approximate tripling component.

For the accelerated scale-6 approximation,

$$P_{N,6}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,6)} P_{6^m N},$$

a chord perturbation at scale  $M = 6^m N$  produces a polygonal perturbation

$$\tilde{P}_M - P_M = \frac{M}{2} \delta_M.$$

Thus

$$\left| \tilde{P}_{N,6}^{[r]} - P_{N,6}^{[r]} \right| \leq \frac{N}{2} \sum_{m=0}^{r-1} |w_m^{(r,6)}| 6^m |\delta_{6^m N}^{(s)}|.$$

Using

$$|\delta_M^{(s)}| \leq A_s M^{-3^s \beta},$$

with  $M = 6^m N$ , gives

$$\left| \tilde{P}_{N,6}^{[r]} - P_{N,6}^{[r]} \right| \leq \frac{A_s}{2} N^{1-3^s \beta} \sum_{m=0}^{r-1} |w_m^{(r,6)}| 6^{m(1-3^s \beta)}.$$

For fixed  $r$ , the finite weighted sum is bounded. If

$$3^s \beta \geq 2r + 1,$$

then

$$N^{1-3^s \beta} \leq N^{-2r}.$$

Therefore the hybrid perturbation is  $O(N^{-2r})$ . Since the exact scale-6 hierarchy satisfies

$$P_{N,6}^{[r]} - \pi = O(N^{-2r}),$$

the perturbed hybrid approximation also satisfies

$$\tilde{P}_{N,6}^{[r]} - \pi = O(N^{-2r}).$$

□

*Remark 7* (Why the Hybrid Case Matters). The numerical experiments indicate that the  $6^m$  family gives the strongest tested hybrid performance. Corollary 3 shows that the hybrid case does not require a fundamentally different perturbation theory. Since the bisection component is already constructible by square roots, the only missing ingredient is a sufficiently accurate constructible tripling component. Thus the hybrid branch reduces to the same tripling seed and cubic-refinement problem as the pure  $3^m$  branch.

## 16 Conclusion

The experiments support a scale-optimized geometric acceleration framework for polygonal approximations to  $\pi$ . The main conclusions are:

1. The polygon sequence has an even-power error expansion.
2. Multi-scale Lagrange extrapolation cancels the first  $r - 1$  even-power error terms.
3. For geometric scale nodes  $b^m$ , the extrapolation weights are uniformly stable.
4. The stability constant is
$$\mathcal{S}_b = \prod_{\ell=1}^{\infty} \frac{1 + b^{-2\ell}}{1 - b^{-2\ell}}.$$
5. Larger scale factors improve stability but do not automatically improve normalized geometric efficiency.
6. The tested data indicate that  $6^m$  is the strongest hybrid branch, while  $3^m$  is the strongest pure tripling branch.
7. The constructible tripling seed problem is solved by the elementary linear seed

$$\tilde{c}_{3N}^{(0)} = \frac{c_N}{3},$$

which has chord error  $O(N^{-3})$ . The Baker cubic refinement step is also compass-and-straightedge constructible and satisfies

$$e_+ = -\frac{7}{384}e^3 + O(e^5).$$

The constructible tripling obstruction is therefore substantially reduced. The scale operators, weight stability, analytic remainder bounds, constructible linear tripling seed, and Baker cubic refinement step are all controlled. The fully geometric tripling hierarchy is obtained by replacing exact cubic inversion for  $c_{3N}$  with the constructible seed

$$\tilde{c}_{3N}^{(0)} = \frac{c_N}{3}$$

followed by finitely many Baker cubic refinements. Since the refinement exponent after  $s$  refinements is  $3^{s+1}$ , the condition

$$3^{s+1} \geq 2r + 1$$

guarantees preservation of the  $r$ -scale accelerated hierarchy. In particular, for  $r = 24$ , three cubic refinements are sufficient.

The central mathematical lesson is that a regular polygon is not merely a slow approximation to the circle. It is a scale-indexed carrier of structured error. Once that structure is exposed, geometric operators can remove it.

The circle is approached not only by refinement, but by cancellation of refinement error.

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